

Gosford High School

Year 11

2009 Preliminary Higher School Certificate Mathematics Extension 1 Assessment Task 2

Time Allowed - 60 minutes
+ 5 minutes reading time

The four questions are of almost equal value

Remember to start each new question on a new page. Work down the page in one column only.

Students must answer questions using a blue/black pen and/or a sharpened B or HB pencil.

Approved scientific calculators may be used

Students need to be aware that

- * 'bald' answers may not gain full marks.
- * untidy and/or poorly organised solutions may not gain full marks.

Linear Function	/12	
Inequalities	/12	
Polynomials	/7	
Algebra	/6	
Polynomials	/13	
Total	/50	

Question 1 (12 Marks)

Marks

- a) A (2, 4), B (5, -3) and C (-2, -6) are the vertices of a triangle on the number plane.
- (i) Find the length of AB 1
 - (ii) Find the gradient of AB 1
 - (iii) Find the mid point of AC 1
 - (iv) What type of triangle is $\triangle ABC$. Fully justify your answer. 3
 - (v) Find the coordinates of D such that ABCD is a parallelogram. 2
- b) Find, in general form, the equation of the line perpendicular to $5x - 3y - 2 = 0$ concurrent with $y = 3x + 2$ and $2x - y = 1$. 4

Question 2 (12 Marks) Begin a New Sheet of Paper

- a) Solve and graph your solution on a number line $\frac{1}{2x-1} \leq 2$ 3
- b) Solve $2x - 1 \leq |x + 4|$ 3
- c) Solve $\frac{3}{(x-2)(x+1)} \geq 0$ 3
- d) Solve $|x+1| + |x-2| = 3$ 3

Question 3 (13 Marks) Begin a New Sheet of Paper**Marks**

- a) Graph $f(x) = (x-3)^2(x+2)(2x-1)$ **2**
- Hence or otherwise, solve $(x-3)^2(x+3)(2x-1) \leq 0$ **2**
- b) Prove that, if $x^4 - x^3 + kx - 4$ has a factor of $(x+1)$, then it also has a factor of $(x-2)$. **3**
- c) Expand and simplify $(2x-3y)(2x+3y) - (2x-3y)^2$ **1**
- d) Factorise $a^2 - b^2 + 2bc - c^2$ **2**
- e) Simplify $\frac{x}{x^3 - y^3} - \frac{1}{x^2 - y^2}$ **3**

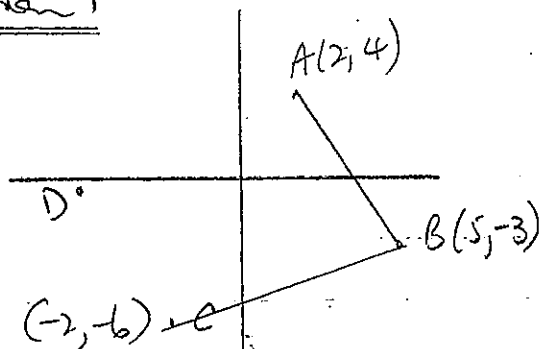
Question 4 (13 Marks) Begin a New Sheet of Paper

- a) When the polynomial $P(x)$ is divided by $x^2 - 1$ the remainder is $3x - 1$.
What is the remainder when $P(x)$ is divided by $x - 1$? Justify your answer. **2**
- b) (i) Factorise completely the polynomial $P(x) = x^3 - 3x + 2$ given that the equation $P(x) = 0$ has a repeated root. **3**
- (ii) The polynomial $Q(x)$ has the form $Q(x) = P(x)(x + a)$ with $P(x)$ as in (i). The constant a is chosen so that $Q(x) \geq 0$ for all real values of x . Find all possible values of a . (Hint: Draw a graph or two.) **1**
- c) If α , β and γ are the roots of $x^3 - 2x^2 + 3x + 7 = 0$, find the values of:
- (i) $\alpha + \beta + \gamma$ **1**
- (ii) $\frac{2}{\alpha} + \frac{2}{\beta} + \frac{2}{\gamma}$ **2**
- (iii) $\alpha^2 + \beta^2 + \gamma^2$ **2**
- (iv) Hence using (i) and (iii) find $\alpha^3 + \beta^3 + \gamma^3$ **2**

Solutions to 2009 Yr 11 Ext 1 task 2

Question 1

a)



i) $d^2 = 3^2 + 7^2 = 9 + 49$
 $d = \sqrt{58}$

ii) $m_1 = \frac{-3-4}{5-2} = -\frac{7}{3}$

iii) $M(0, -1)$

iv) For BC: $d^2 = 7^2 + 3^2$
 $d = \sqrt{58}$
 $m_2 = \frac{-3+6}{5+2} = \frac{3}{7}$

$BC = AB$ and $BC \perp AB$

so $m_1 m_2 = -1$

$\therefore \triangle ABC$ is isosceles & right angled

v) $D(-3, -1)$

b) $2x - (3x + 2) = 1$

$2x - 3x - 2 = 1 \Rightarrow x = -3$
 $y = -7$

$m_1 = \frac{5}{3} \therefore m_2 = -\frac{3}{5}$

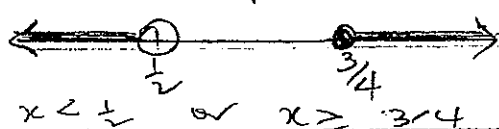
$\therefore y + 7 = -\frac{3}{5}(x + 3)$

$5y + 35 = -3x - 9$

$3x + 5y + 44 = 0$

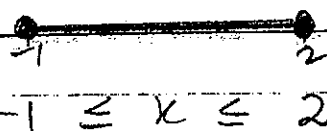
Question 2

a) $x > \frac{1}{2}$ } $x < \frac{1}{2}$
 $1 \leq 4x - 2$ } $1 \geq 4x - 2$
 $\frac{3}{4} \leq x$ } $\frac{3}{4} \geq x$



b) $2x - 1 \leq |x + 4|$
 For $x \geq -4$ } For $x < -4$
 $2x - 1 \leq x + 4$ } $2x - 1 \leq -x - 4$
 $x \leq 5$ } $3x \leq -3$
 $x \leq -1$

d) $|x + 1| + |x - 2| = 3$
 Distance of x from -1 plus distance of x from 2 equals 3 .



c) Critical points $x = 2, -1$
 No solutions to $3 = 0$



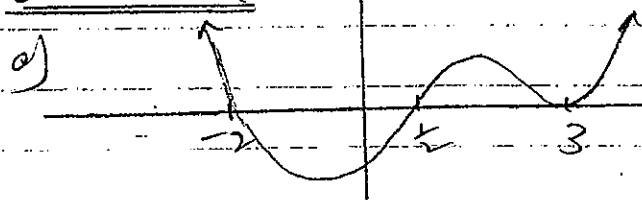
Test $x = 0$ $\frac{3}{-2} \neq 0$

Test $x = 3$ $\frac{3}{1 \times 4} \geq \checkmark$

Test $x = -2$ $\frac{3}{(-4)(-1)} \geq \checkmark$

$\therefore x < -1$ or $x \geq 2$

Question 3



$-2 \leq x \leq \frac{1}{2}$ or $x = 3$

b) $P(-1) = 1 + 1 - k - 4 = 0$
 $k = -2$

$\therefore P(x) = x^4 - x^3 - 2x - 4$

$P(2) = 16 - 8 - 4 - 4 = 0$

$\therefore (x - 2)$ is a factor

$$c) 4x^2 - 9y^2 - 4x^2 + 12xy - 9y^2 \\ = 12xy - 18y^2$$

$$d) a^2 - (b^2 - 2bc + c^2) \\ a^2 - (b-c)^2 \\ (a+b-c)(a-b+c)$$

$$e) x \\ (x-y)(x^2+xy+y^2) = \frac{1}{(x-y)(x+y)}$$

$$\frac{x(x+y)}{(x-y)(x+y)(x^2+xy+y^2)}$$

$$\frac{x^2+xy - x^2-xy-y^2}{(x-y)(x+y)(x^2+xy+y^2)}$$

$$= \frac{-y^2}{(x-y)(x+y)(x^2+xy+y^2)}$$

Question 4

$$a) P(x) = (x-1)Q(x) + 3x-1 \\ \therefore P(1) = 0 + 3 \times 1 - 1 \\ P(1) = 2 \quad \text{Remainder is 2.}$$

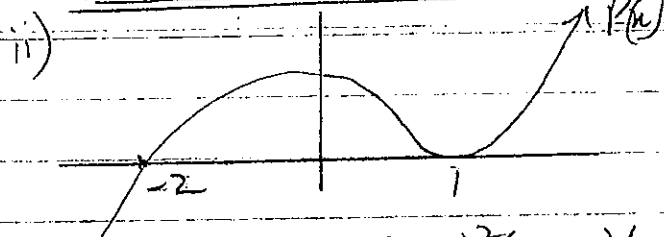
$$b) P(1) = 1 - 3 + 2 = 0 \\ x^2 + x - 2$$

$$\begin{array}{r} x-1 \overline{) x^3 + 0x^2 - 3x + 2} \\ \underline{-x^3 + x^2} \\ x^2 - 3x + 2 \\ \underline{-x^2 + x} \\ -2x + 2 \\ \underline{-2x + 2} \\ 0 \end{array}$$

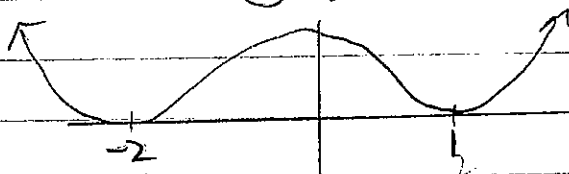
$$P(x) = (x-1)(x^2+x-2) \\ = (x-1)(x-1)(x+2)$$

$$P(x) = (x-1)^2(x+2)$$

$$P(x) = (x-1)^2(x+2)$$



Now $Q(x) = (x-1)^2(x+2)(x+a)$
For $Q(x) \geq 0$ for all x



is only choice.

$$Q(x) = (x-1)^2(x+2)^2 \\ a = 2 \text{ only}$$

$$c) i) \alpha + \beta + \gamma = \frac{-b}{a} = 2$$

$$ii) \frac{2}{\alpha} + \frac{2}{\beta} + \frac{2}{\gamma} = \frac{2(\beta\gamma + \alpha\gamma + \alpha\beta)}{2\beta\gamma} \\ = \frac{2 \times 3}{7} = \frac{-6}{7}$$

$$iii) \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma) \\ = 2^2 - 2 \cdot 3 = -2$$

$$iv) \text{Sub } x = \alpha \text{ into eqn} \\ \alpha^3 - 2\alpha^2 + 3\alpha + 7 = 0$$

$$\text{or } \alpha^3 = 2\alpha^2 - 3\alpha - 7 \quad (1)$$

$$\text{Sim } \beta^3 = 2\beta^2 - 3\beta - 7 \quad (2)$$

$$\text{Sim } \gamma^3 = 2\gamma^2 - 3\gamma - 7 \quad (3)$$

$$(1) + (2) + (3) \\ \alpha^3 + \beta^3 + \gamma^3 = 2(\alpha^2 + \beta^2 + \gamma^2) - 3(\alpha + \beta + \gamma) - 21$$

$$= 2(-2) - 3(2) - 21$$

$$= -4 - 6 - 21$$

$$= -31$$